A conditional assembly from Theorem 3.11 to Corollary 3.12: decomposition of the scalar layer, a semantic principle, and isolation of the value-group profile switch

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Abstract

We report a *conditional assembly* of the passage from Theorem 3.11 to Corollary 3.12 in inter-universal Teichmüller theory (IUT), expanding the seams that the source treats by abbreviation conventions and citations to numbered results. This paper does not claim an unconditional proof of Corollary 3.12 or of the ABC conjecture. The claim has the following form: accept the numbered statements of IUT I–IV as black boxes, take Mochizuki's non-IUT papers ([FrdI], [FrdII], [EtTh], [AbsTopIII], etc.) and classical results as known inputs, accept a single semantic principle S (prime-strip isomorphisms are isomorphisms of collections of data intertwining all constituents, and compatibility is read as commutative squares), and moreover accept, as an explicit input, the central value-group switch **H*val** (hull compatibility of the switch from Ψ_{LGP} to Ψ_q); then each step of the passage is filled by named grounds. The scale-preservation problem is decomposed into local cancellation, chart transport (distortion = 1), an explicit formula for the mono-analytic correction, distortion of evaluation isomorphisms, and mutual compatibility of cyclotomic rigidities. Of these, the *scalar layer* — chart calibration, generator scalars and transport distortion — is supplied at the statement level using material outside the cited range; this is a part that was not derivable in the cited theory of [U26]. An external adjudication has however settled that the loop argument of Lemma 8.1 proves only identity of the chart calibration for the two routes, not equality of pilot profiles. The value-group switch therefore remains unresolved, as the core of strong (SHE)/(9-1). Sufficiency of the assembly is also machine-checked in Lean 4 (the capstone `cor312_of_assembly`, standard three axioms only in the recorded selected build). What remains is the IUT black box itself, the value-group switch H*val, inhabitation of Principle S in the actual construction (frame fidelity), and independent re-verification of this paper. The correspondence with preceding reductions (the Scholze–Stix critique, Project LANA's (9-1) and `xi_iff`) is also made explicit.

The difference from the author's preceding work [U26], *Where Theorem 3.11 sits*, is stated at the outset. What [U26] proved is negative: from the statements actually cited in the proof of Corollary 3.12 alone, Corollary 3.12 does not follow (a countermodel exists). The present paper works in the opposite direction: using the content of the actual construction outside the cited range, it assembles the passage at the statement level (the level at which numbered statements are used as black boxes and the seams between them are filled by expanding definitions). In the author's view, many of the seams written out here may correspond to parts that were present in the IUT author's mind but were not written down, inside the proof of Corollary 3.12, as citable

statements; the work of this paper may be read as an attempt to supply that unwritten layer. This is the author's interpretation, not a claim about the IUT author's intentions.

Keywords: inter-universal Teichmüller theory, Corollary 3.12, scale preservation, multiradial representation, completeness, formalization (Lean), cyclotomic rigidity.

Note added in version 1.0.2. Figure 1 and its caption are redrawn. Version 1.0.1 had added a caution that the diagram was misrendered (overlapping labels); that caution is removed. The caption now states the claimed relations explicitly, and those concerning the assembly range B are checked in Lean (`VennRegionB.lean`, added to the supplementary files). The Lean counterpart of H*val, given as `hmem` alone in the table of §1, is corrected to `hmem` and `hlocal`, with one sentence added in §2. The description of the unpublished working draft [W] is reworded. No result or conclusion is changed.

Note added in version 1.0.3. In the supplementary Lean files, the comments are rewritten in English and the root namespace of the excerpts is renamed. No statement, proof or result is changed.

Claim and what remains, at the outset

The claim of this paper is that what have been treated as "walls" — Corollary 3.12, the height inequality, LANA's (9-1) — have been reduced to the following list of remaining items. If all of them are established, Corollary 3.12 follows, and through the machinery of IUT IV an ABC-type height inequality follows.

In relation to the preceding work: of what was underivable in [U26], the *scalar layer* (chart calibration, generator scalars, transport distortion) is supplied in this paper, and what remains at the centre is the *profile layer*. Supplying the scalar layer does not by itself mean that the aggregate defect Δ vanishes. The step that carries a raw Θ-pilot profile to a q-pilot representation remains as item R2V below.

- **(R1) The IUT black box.** Correctness of the numbered statements of IUT I–IV, including Theorem 3.11. Not verified here.
- **(R2V) The value-group switch H*val.** The switch $\Psi_{\mathrm{LGP}} \to \Psi_q$ is compatible with the holomorphic hull, and an actual q-pilot representation is obtained from the $\{j^2 u_{\underline{v}}\}_j$-type Θ-pilot profile. Lemma 8.1 proves only that the two routes give the same Φ^{rlf} chart calibration; Lemma 8.2 treats only the unit side. This is the core of strong (SHE)/(9-1) (§8).
- **(R2) Inhabitation of Principle S in the actual construction.** Each isomorphism that the source calls a prime-strip isomorphism (evaluation, Kummer, links), also in situations where the same object is identified in other ways, satisfies the two commutative squares and non-degeneracy as a morphism in the sense of Definition 5.2 (in Lean: inhabitation by the actual objects of `TwoSquareFVdashIso` = frame fidelity). The phenomena already reproduced in models and the precise transfer problem to the actual construction are detailed in §3.1.
- **(R3) Independent re-verification of this paper.** Not a mathematical hypothesis but work not yet done, including completeness of the enumeration of seams.
- **(R4) Only for explicit ABC constants.** Numerical values of $B^*_{\mathcal{K}}$ for concrete $\mathcal{K}_V$ (computable from the definition; a finite example is in §11) and treatment of the coefficient case $(\ell, d_{\mathrm{mod}}) = (5, 1)$. Not needed for Corollary 3.12 itself.

Conversely, the content of the assembly in this paper is that no input other than these is needed. Figure 1 updates the Venn diagram of [U26] to the present state.

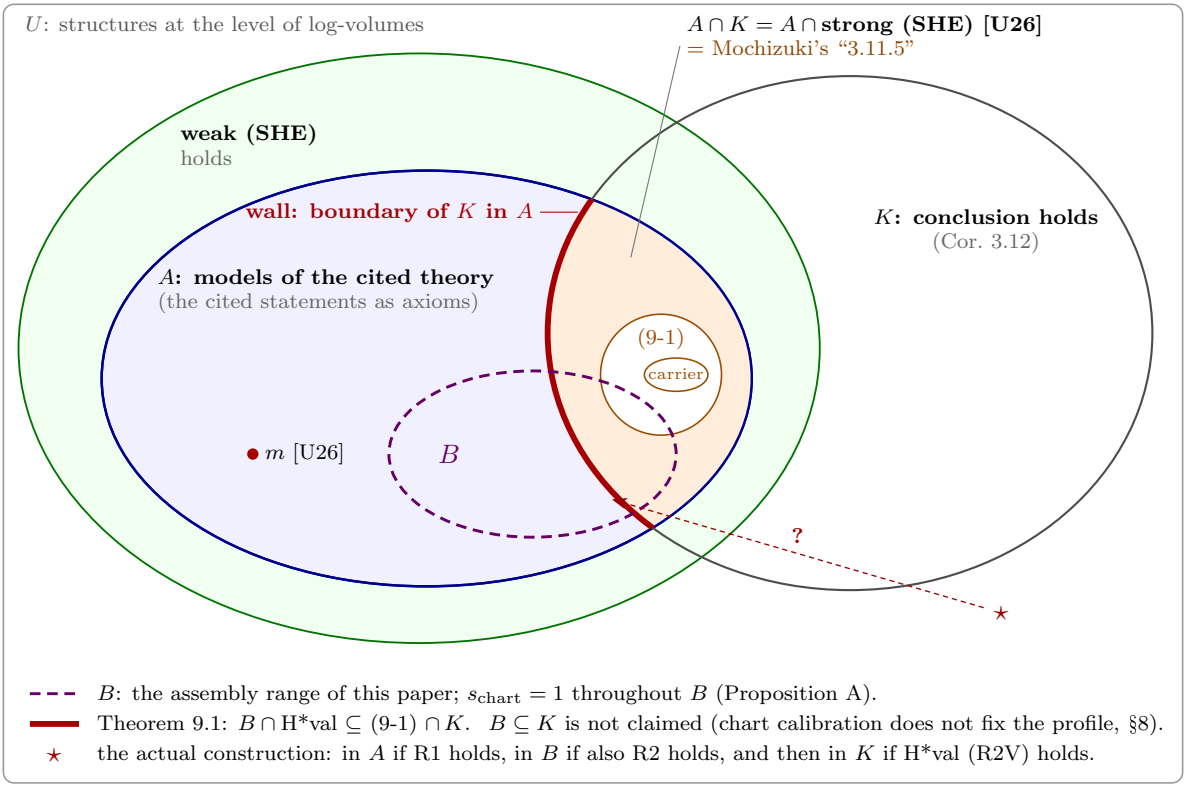


Figure 1: Updated Venn diagram. As in [U26], each statement is drawn as the class of structures in which it holds; sizes, shapes and positions are schematic and measure nothing. The relations claimed are exactly the following. (a) A lies inside weak (SHE), and A contains a model m outside K [U26]. (b) Inside A, K coincides with strong (SHE) in the value reading; the carrier class lies inside (9-1), and (9-1) inside K [U26]; $A \cap K$ is the locus of Mochizuki's "3.11.5" (§10). (c) B, the assembly range of this paper, is the class of models of the cited theory that also satisfy the inputs of Theorem 9.1 other than H*val; it lies inside A, and $s_{\text{chart}} = 1$ throughout B (Proposition A). (d) $B \cap$ H*val $\subseteq$ (9-1) $\cap K$ (Theorem 9.1 and §10). $B \subseteq K$ is not claimed: B is not a range where (9-1) has been solved, but the range in which (9-1) has been isolated as the value-group switch H*val (R2V). In the abstract frame of the Lean capstone (§2), (c), (d) and the inclusion (9-1) $\subseteq K$ are proved in Lean, and B meets both sides of the wall, so that $B \not\subseteq K$ in that frame (by the profile countermodel of [U26c]); this holds whether H*val is read as `hmem`, as `hlocal`, or as both (`VennRegionB.lean`). (e) The star is the actual construction. The IUT statements as a whole (R1) are a premise outside the cited range defining A; inhabitation of Principle S by the actual objects is R2; whether the star lies in K is what remains of the wall. Beyond these, no inclusion and no non-emptiness of a region is claimed.

1 Introduction and position

Contrast with the preceding work. [U26] formalized the range of statements cited in the proof of Corollary 3.12 and showed, by a countermodel, that Corollary 3.12 does not follow from that

range alone. The shortages localized there were, in the terminology of the time, an input fixing the scale s, and uniformity in (IPL). In this paper the former is further split into the *scalar layer* (chart calibration, generator scalars, transport distortion) and the *profile layer*, which concerns the arrangement of value-group components. As a sequel, this paper identifies step by step where in the actual construction of the source the scalar layer is supplied from, isolates the value-group switch at the centre of the profile layer as a named remainder H*val, and assembles its surroundings at the statement level. The two are therefore not in conflict: the countermodel of [U26] remains valid, and the assembly here explicitly uses content outside the cited range (explicit formulae, uniqueness characterizations, expansions of definitions).

The author's interpretation. The seams expanded here — the explicit formula for the mono-analytic correction, computation of distortion of evaluation isomorphisms, mutual compatibility of cyclotomic rigidities, etc. — are treated in the source by abbreviation conventions, one-line proofs, and remarks. In the present author's view, these may correspond to parts that were present in the IUT author's mind but were not written down, inside the proof of Corollary 3.12, as citable statements; the work here resembles an attempt to supply that layer. Whether this interpretation is apt is not part of the claims of this paper, and whether the present assembly is correct as such a supplement awaits independent re-verification.

The author's purpose is understanding of IUT, and this paper is work of verification that reads through the IUT papers. The mathematical content is attributed to IUT I–IV of Shinichi Mochizuki, the founder of IUT; the contribution here is limited to making the seams explicit and decomposing them into a verifiable form. This paper has not received confirmation or approval from Professor Mochizuki.

Corollary 3.12 of IUT III, which derives the log-volume inequality $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$ from the multiradial representation of Theorem 3.11, has been at the centre of the controversy since 2018. In [U26] we showed that axiomatizing only the statements cited by Corollary 3.12 admits a countermodel, and that an input fixing the scale s and uniformity in (IPL) do not exist inside the cited range. In the sequel [U26b] the problem was formulated as completeness for forgetful maps, and the sufficient condition for ABC was reduced to the aggregate defect Δ (v1.1.3 clarifies that Δ is not a chart multiplier itself but an aggregate quantity that may depend on the profile). The main countermodel of [U26] is not a model obtained by arbitrarily stretching a chart. It is a profile model with $\theta_j = B(j, 29j^2)$, $q_j = B(j, 29)$, and $s = 3/14 = 12/(\ell(\ell+1))$ is an aggregate scalar forced as the mean of that j^2 profile. This paper is a sequel: using the content of the actual construction outside the cited range, it identifies where the scalar layer is supplied from, step by step, and isolates the profile layer that still remains.

Priority for the reduction belongs to the Scholze–Stix critique and to Project LANA. In particular, LANA's (9-1) is the central sentence corresponding to the value-group switch H*val isolated in §8 here, and this paper does not claim to have solved it. The dictionary is given in §19 of the working draft [W]. The contribution here is not a new mathematical idea but: (i) an enumeration of seams with named grounds for each step; (ii) a rating that includes the history of corrections of overstatements by external cross-checking; (iii) an explicit finite list of remaining dependencies, including H*val and in-situ inhabitation of Principle S; and (iv) machine verification of sufficiency in Lean. The detailed proof material is in the working draft [W] (unpublished, in Japanese); this note gives a summary and its position. The passage corresponding to Lemma 13.5 of [W] v0.14 still needs revision in line with the downgrade in §8 here.

What this paper does not claim. An unconditional proof of Corollary 3.12 or of the ABC conjecture; verification of the correctness of IUT itself, including Theorem 3.11; or a rebuttal of the Scholze–Stix critique. This paper is a conditional assembly — "if the listed inputs are accepted,

the passage assembles" — together with a complete explicit list of those inputs.

2 Reduction to defect form and the Lean capstone

Fix notation. Let $v_{\mathrm{out}} = \mu^{\log}(T_\lambda)$ be the global readout of the Θ-link image T_λ after transport to a common q-side chart, and let $v_Q = -|\log(\underline{q})|$ be the volume of the input q-pilot. The scale $s \in \mathbb{R}_{>0}$ is defined by $v_Q = s \cdot v_{\mathrm{out}}$, and the defect by $\Delta = (s-1)v_{\mathrm{out}}$. The old notation $\mathrm{vol}\,\theta$ for the raw volume before transport is not used here, since the readout side becomes ambiguous. The sufficient condition for the output hull H reduces to two sentences: the input sentence is exact scale preservation $s = 1$ (i.e. $\Delta = 0$); the output sentence is that the single global object T_λ belongs to the family of possible images and $v_{\mathrm{out}} \le \mathrm{vol}\, H$. As for signs: if $\mathrm{vol}\, H \ge 0$ then $v_Q < 0 \le \mathrm{vol}\, H$ and the conclusion is trivial, so below $s > 0$ is discussed on the nontrivial region $\mathrm{vol}\, H < 0$, in particular $v_{\mathrm{out}} < 0$. This reduction and defect algebra are verified in Lean 4, with the standard three axioms only.

$$v_Q = s \cdot v_{\mathrm{out}}, \qquad \Delta = (s-1)v_{\mathrm{out}}, \qquad \Delta = 0 \iff s = 1 \ (v_{\mathrm{out}} \neq 0), \qquad \text{conclusion: } v_Q \le \mathrm{vol}\, H.$$

The capstone `cor312_of_assembly` constructed in this work (Lean; PASS on a selected closure) does not receive the input sentence $s = 1$ as a hypothesis; instead it derives it from place-by-place equality of corrected volumes and a decomposition into a common weighted sum. The remaining hypotheses are made explicit as three items — volume decomposition (transport and frame fidelity), corrected place-by-place equality, and the membership sentence — and are not axioms. The Lean membership hypothesis `hmem` is $T \in \mathrm{Fam}$ for a single global object T, not place-by-place membership; only the numerical equality `hlocal` is place-by-place. The value-group switch H*val (R2V) enters both: the proof of Theorem 9.1 uses it for the membership `hmem` and, through the profile, for `hlocal`. Variants comparing on the tensor-power side, `cor312_of_assembly_scaled`, and deriving $s = 1$ itself, `scale_eq_one_of_assembly`, are also included. For the computational core of local cancellation, equality of pilot degrees for a Tate model pair is proved in Lean. Identification of the model pair with the actual construction (frame fidelity) remains work on the Lean side.

3 Semantic Principle S and the ρ-square

Semantic Principle S [foundation]. Isomorphisms of prime-strips, as isomorphisms of collections of data, intertwine all constituents (in particular $\{\rho_{\underline{v}}\}$), and "compatibility" of two displayed isomorphisms is read as a commutative square. Dependence on identifications in the scalar layer and in the unit-side assembly is collected into this single principle. The value-group switch H*val is separated as an independent input in §8.

Lemma 3.1 (ρ is a constituent) [source]. In the displays $(\mathcal{C}^{\Vdash}, \mathrm{Prime} \simeq \underline{\mathbb{V}}, \{\mathcal{F}^{\vdash}_{\underline{v}}\}, \{\rho_{\underline{v}}\})$ of $\mathfrak{F}^{\Vdash}_{\mathrm{mod}}$ and $\mathfrak{F}^{\Vdash}_{\mathrm{tht}}$, the families $\{\rho_{\underline{v}}\}$, $\{\rho^{\Theta}_{\underline{v}}\}$ occur as constituents of prime-strip data. *Proof.* By the display at the end of Example 3.5 (ii). The general form is as in Definition 5.2 (iv). □

Lemma 3.2 (the ρ-square) [source] + expansion of definitions. A morphism of $\mathcal{F}^{\Vdash}$-prime-strips (an isomorphism of collections of data in the sense of Definition 5.2 (iv)) intertwines the constituents $\{\rho_{\underline{v}}\}$. That is, for the induced isomorphisms on Φ, Φ^{rlf}, for each $\underline{v}$ the square $\Phi^{\mathrm{rlf}}(\beta_{\underline{v}}) \circ \rho_{\underline{v}} = \rho'_{\underline{v}} \circ \Phi(\alpha)$ commutes. *Proof.* Definition 5.2 (iv) defines a prime-strip as the tuple $(\mathcal{C}^{\Vdash}, \mathrm{Prime}(-) \xrightarrow{\sim} \underline{\mathbb{V}}, \mathfrak{F}^{\vdash}, \{\rho_{\underline{v}}\})$ and its morphisms as "isomorphisms of collections of data as above".

The components of such an isomorphism are a category equivalence α, compatibility with the prime bijection, a family of local prime-strip isomorphisms $\beta_{\underline{v}}$, and the ρ-square; the square is included as part of the definition of an isomorphism of collections of data. Example 3.5 (ii) states, in the model case, that $\rho_{\underline{v}}$ and $\rho_{\underline{v}}^{\Theta}$ are compatible with the natural isomorphism and that this fact can be expressed as a natural isomorphism of collections of data $\mathfrak{F}_{\mathrm{mod}}^{\Vdash} \xrightarrow{\sim} \mathfrak{F}_{\mathrm{tht}}^{\Vdash}$. □

This lemma itself is almost immediate as an expansion of definitions. The issue lies at the level of application: whether each isomorphism that the source calls a prime-strip isomorphism (evaluation isomorphisms, Kummer isomorphisms, links) remains a morphism in the sense of Definition 5.2 also in situations where the same object is identified in other ways. Here each application is used only where a numbered statement of the source explicitly states that it is a prime-strip isomorphism (§7).

3.1. Breakdown of R2: what the models show, and the transfer problem

R2 above is not about whether Lemma 3.2 itself holds. Lemma 3.2 is a consequence once objects are given as tuples of Definition 5.2 and morphisms as isomorphisms of those tuples. What remains is whether the construction in the source actually supplies objects and morphisms of that type, and whether the required properties continue to hold simultaneously under the identifications stacked in the proof of Corollary 3.12. Stated precisely, the unresolved sentence is the following.

R2 (precise unresolved sentence). For each prime-strip built from the Hodge theatres of the source, and for each evaluation / Kummer / $\mathfrak{log}$- or Θ-link / permutation isomorphism, the correspondences of primes, local objects, Φ/Φ^{rlf}, ρ and pilot objects can be chosen coherently so that, under that choice, the prime-label square and the ρ-naturality square hold and the distinctions between the pilots being compared are not collapsed. Moreover, the local component isomorphisms must actually be induced by the global and local isomorphisms.

What has been shown at the level of models and types is limited to the following four items. First, the bare shape of a quadruple does not automatically come with the two squares, and the two squares are independent; finite models with a wrong prime correspondence and a wrong ρ correspondence show this. Second, shape and commutativity alone do not exclude degeneracy: the closed unit model collapsing everything to $\mathbf{Z}/1$ satisfies both squares but is not non-degenerate (Lean: `source_shape_and_squares_do_not_exclude_closed_unit`). Third, in a finite synchronized model and in a real-valued model by weight scaling on Multiplicative $\mathbb{R}$, both squares and non-degeneracy hold simultaneously; the type is not an inconsistent demand. Fourth, in a Tate model pair, for generators built from an actual local Tate parameter and theta roots, although the unnormalized coefficients differ as $1/(2\ell)$ and 1, the normalized generators agree on the same Tate coordinate, the theta-side generator agrees with the negative logarithmic readout of the actual theta-root class, and the pilot degrees agree (Lean: `coe_gen_thetaTateStrip_eq_readout`, `pilot_degree_thetaTateStrip_eq`). Thus the phenomenon of local cancellation itself is reproduced in a model whose input is actual Tate data, not arbitrary positive reals.

The transfer is nevertheless not shown. The prime-strips of the Tate model are constructed inside our `PStrip` interface from given families of generators and charts; they are not the Frobenioids of the source themselves. The existing prime-strip skeleton is also a partial interface: it does not yet state that `globalLocIso`/`localRlfIso` are induced by the global and local isomorphisms, nor coherence as a category equivalence. What is needed is not merely that "a similar phenomenon occurs somewhere", but that the source's ρ^{Θ}, the theta values of [EtTh], and the Kummer classes continue to point to **the same generators and the same pilot objects** after stacked identifications, that the squares hold simultaneously for the whole composites used at each step of the proof,

and that the relevant distinctions of degrees and regions are not collapsed. Squares holding locally for separately chosen representatives do not suffice; the differences arising on composition must lie within the permitted indeterminacies. This is the content of R2. The value-group switch H*val (R2V) of §8 is not transfer to a Lean model but hull compatibility of $\Psi_{\mathrm{LGP}} \to \Psi_q$ inside the source construction itself, and is distinguished from R2. What remains at the centre is the conjunction of these two.

4 Local cancellation

Fix a bad prime $\underline{v} \in \underline{\mathbb{V}}^{\mathrm{bad}}$ and take as a common coordinate the Tate coordinate $u = -\log|q_{\underline{v}}| > 0$. The normalization that a generator corresponds to valuation 1 is by [AbsAnab] Proposition 1.2.1 (iv) (KNOWN-1: the Frobenius element is tracked by the normalized valuation; the same normalization is used in [AbsTopIII] Corollary 1.10 (b)).

Lemma 4.1 (local cancellation by generator preservation) [assembly; computational core Lean-checked]. Define the value-group member scalar $s_{\underline{v}}$, in a common Φ^{rlf} chart, by "the θ-side generator normalized by $\rho^{\Theta} = s_{\underline{v}} \times$ (the normalized generator on the q-side)". Then $s_{\underline{v}} = 1$. *Proof.* The generator on the mod side is the 2ℓ-th root $\underline{\underline{q}}_{\underline{v}} = q_{\underline{v}}^{1/2\ell}$ of the q-parameter, and its Φ-logarithm gives the generator (Example 3.2). Its value in the realified common chart is u. On the theta side, by the formula of [EtTh] Proposition 1.4 (ii), the value of $\underline{\underline{\Theta}}_{\underline{v}}$ at $\sqrt{-q_{\underline{v}}}$ equals $q_{\underline{v}}^{1/2\ell}$. The unnormalized theta-side Φ-generator has the form $\log(p_{\underline{v}}) \cdot \log(\underline{\underline{\Theta}})$, and the ρ^{Θ} normalization appears as division by the ratio $\log\Theta/\log q$ of this generator to the mod-side generator (the formula of Example 3.5 (ii)). Hence the normalized theta-side generator also has value u. By Lemma 3.2 this generator correspondence intertwines with transport by prime-strip isomorphisms, so the defining equality becomes $u = s_{\underline{v}} \cdot u$, and $u \neq 0$ gives $s_{\underline{v}} = 1$. The computational core is checked in Lean by `rho_square_cancellation`, etc. □

Lemma 4.2 (preservation of per-step intrinsic chart ratios) [assembly]. As long as each transport step is read in its own intrinsic generator chart, the ratio 1 of Lemma 4.1 is preserved. The steps are (T1) construction of local LGP monoids (Proposition 3.4 (ii)), (T2) global LGP prime-strips (Proposition 3.7 (iii)), and (T3) realification and product embedding (Proposition 3.7 (v)). This lemma does not assert descent to a single $\mu^{\log}$. *Proof.* (T1) is a functorial algorithm on $\mathfrak{log}$-link input, built by images and pullbacks, hence preserves ratios of generators. (T2) is a displayed tuple of prime-strips, to which Lemma 3.2 applies directly. (T3) is given by a commutative diagram of categories and vertical isomorphisms, built to be compatible with $\{\rho_{\mathrm{LGP},\underline{v}}\}$ and $\{\rho_{\mathfrak{lgp},\underline{v}}\}$. The LGP reading of Gaussian citations (the remainder (α')) is treated, in the range used here, by checking that the dependence is only on vertical or single-theatre input ([W], §12). □

5 Transport and Proposition A

Lemma 5.1 (fixed-family compatibility: transport from charts to $\mu^{\log}$ and distortion) [assembly]. The intrinsic chart ratio 1 of Lemma 4.2 descends, through the normalization of Proposition 3.9, to a degree ratio 1 for a single procession normalization $\mu^{\log}$. Here [AbsTopIII] Propositions 5.7, 5.8 and Corollary 5.10 are taken as known input under the policy of §12. The claim is that transport distortion for a *fixed* family of objects is 1; it is not equality of degrees

between different objects. *Proof.* (a) *Anchor of packet normalization.* Proposition 3.9 (i) puts the log-volume of the local holomorphic integral structure $\mathcal{O}$ equal to zero and normalizes so that multiplication by $p_{\underline{v}}$ corresponds to adding $-\log p_{\underline{v}}$. The weights are explicit in Remark 3.1.1 (ii) and keep this anchor uniform across primes and extensions. The normalization on the generator-chart side (KNOWN-1) is fixed to the same local data. Capsule averaging is defined as the same operation on objects on both sides; that its numerical effect differs by object is not distortion. (b) *Mono-analytic correction.* The correction in Proposition 3.9 (ii) is, by Remark 3.9.1 (i), the log-volume of the mono-analytic integral structure $\mathcal{I}_i = (p^*_{v_{\mathbb{Q}}})^{-1} \log_{k_i}(\mathcal{O}^\times_{k_i})$, and from the formula of [AbsTopIII] Proposition 5.8 (iii) it computes to $c = -(f+m)\log p + ef \log p^*$ (the Archimedean side is $\log \pi$), where e is the ramification index, f the residue degree, $|\mu_{p^\infty}(k_i)| = p^m$, $p^* = p$ for p odd and 4 for $p = 2$; this is the local quantity before normalization by $ef = [k_i : \mathbb{Q}_p]$. In the unramified case and for $\mathbb{Q}_2$, $c = 0$ was also confirmed by classical computation. Quantitative evaluation of this deviation is the subject of IUT IV: a matter of constants, not of transport of ratios. (c) $\mathfrak{log}$*-link transport.* By the compatibilities of Proposition 3.9 (iv) and Propositions 1.2 (iii), 1.3 (iii). (d) *Regions.* Under the caveat of Remark 3.1.1 (iii), the regions here lie on the product-region side. As a remainder, the step where Proposition 1.2 (iii) reduces to [AbsTopIII] Proposition 5.7 (c) and Corollary 5.10 (ii) is accepted as a statement of the IUT main text. □

Proposition A (chart/transport scale preservation, global form) [assembly]. Under Lemmas 3.2, 5.1 and single-scalarity ([FrdI] Theorem 6.4 (ii), a known input), the global chart degree of a concentrated member is 1 and the chart defect is $\Delta_{\mathrm{chart}} = 0$ (i.e. $s_{\mathrm{chart}} = 1$). *Proof.* At each bad prime the member scalar is 1 by Lemma 4.1, and at good primes it is trivially 1. By Lemma 3.2, $\{\rho_{\underline{v}}\}$ is transported as a constituent of prime-strip data, and by Lemma 5.1 the ratio descends to a single $\mu^{\log}$. Since the global scalar is single by [FrdI] Theorem 6.4 (ii), $s_{\mathrm{chart}} = 1$, i.e. $\Delta_{\mathrm{chart}} = 0$. □

Note: Δ_{chart} in Proposition A is not the aggregate defect Δ of §2. Proposition A is a claim of the scalar layer, concerning transport of a fixed family of objects; it does not assert that the j^2 profile of a Θ-pilot and the constant profile of a q-pilot agree as objects. That switch is H*val in §8, and only by combining Proposition A with the explicit hypothesis H*val does Theorem 9.1 obtain the input-side reading.

6 The unit part and Lemma B

Lemma 6.1 (γ': compatibility as a typed commutative square) [assembly]. The difference is read as the type-consistent auto-poly-isomorphism $\delta^{\vdash\times\boldsymbol{\mu}} := (\Phi^{\vdash\times\boldsymbol{\mu}}_{0,\circ})^{-1} \circ \lambda^{\vdash\times\boldsymbol{\mu}}_\circ$ (the naive composite $\Phi_0^{-1} \circ \lambda$ is not used, as its types do not match). Its components lie in the orbit of (Ind1)(Ind2)(Ind3). *Proof. Anchors of the proof.* Anchor 1: the end of Theorem 3.11 (i) states explicitly that the construction algorithm for ${}^\circ R^{\mathrm{LGP}}$ is functorial with respect to procession isomorphisms. Anchor 2: the morphism components of Corollary 2.3 are necessarily compatible with the poly-isomorphism of radial data, and the functor is full. Anchor 3: Theorem 3.11 (iii)(a)(b) is in the form asserting compatibility of two displayed full poly-isomorphisms. Anchor 4: Remark 3.11.4 (ii) states that no further indeterminacy can occur, citing the general theory of [FrdII] Definition 2.1 (ii) as grounds (verbatim-checked). A component arising from $\mathrm{Aut}(G)$ is trivial on $\mathcal{O}^{\times\boldsymbol{\mu}}(G) \twoheadrightarrow \mathbb{Q}_p$, does not move scalars, and is absorbed as a prime-strip automorphism on the theatre side. On the other hand, a component on the Ism side is the definition of (Ind2) itself. Hence movable differences are limited

to the Ism side, but they are manifested as indeterminacy and are not identified with absorption on the $\mathrm{Aut}(G)$ side. □

Lemma B (decomposition) [assembly]. Under the value-group switch **H*val** (§8), the global representation of the link image T_λ belongs to the upper family of the holomorphic hull ${}^{1,\circ}\overline{\mathcal{U}}$ and satisfies $\mu^{\log}(T_\lambda) \leq -|\log(\underline{\Theta})|$. Here (Ind1)(Ind2)(Ind3) bear only the absorption of unit-side differences; it is not asserted that $T_\lambda = \iota(T_0)$ is obtained by a single (Ind) variation ι. The switch itself, from a $\{j^2 u_{\underline{v}}\}_j$-type profile to an actual q-pilot representation, is the content of H*val. Membership and volume comparison concern a single global object and a single global number; the indices $(j, v_{\mathbb{Q}})$ are a component display of containers, not place-by-place assertions. *Sketch of proof.*

By H*val, the switch from Ψ_{LGP} to Ψ_q is compatible with the holomorphic hull, and the global representation of the actual output enters the upper family of the hull as a q-pilot representation. The remaining difference is the typed unit-part difference $\delta^{\vdash\times\boldsymbol{\mu}}$, absorbed as an (Ind) variation by Lemma 6.1. Hence the global representation belongs to ${}^{1,\circ}\overline{\mathcal{U}}$, and the volume bound above follows from the definition of the hull. The detailed step decomposition is in [W], §4. □

7 The evaluation side: numerical bookkeeping, assembly, distortion

Lemma 7.1 (numerical identity for (Ob3)) [assembly]. The step that Remark 3.9.5 (vii) (Ob3) leaves as "routine details to the reader" can be treated as numerical bookkeeping of degrees before and after applying $\det_{\otimes M}$. This numerical identity alone is not an object-level comparison; it can be used as a comparison only under Lemmas 7.2, 7.4 and the value-group switch H*val of §8.

Lemma 7.2 (well-definedness of the assembly) [assembly]. By the object-level identification for (Ob4) (the output of $\det_{\otimes M}$ gives an object of the target prime-strip via the naive Frobenius functor of [FrdI] Proposition 2.1 (i)) and single-scalarity ([FrdI] Theorem 6.4 (ii)), assembly of local data into a single global degree is well-defined. This lemma does not assert realization of the value-group switch itself. *Sketch of proof.* (Ob4-2) itself places the relevant local Frobenioids in the model class of [FrdI] Theorem 5.2 (ii), so Theorem 6.4 applies. By Propositions 2.1 (i)(ii), the non-tensor-power version determines the tensor-power version and degrees are multiplied uniformly by M; since the comparison is made on the tensor-power side, the twist indeterminacy of reconstruction ((Ob4-2)(Ob4-3)) does not touch the numerical comparison. □

Lemma 7.3 (relative compactness) [assembly]. Possible images are contained in a tensor packet of log-shells and are bounded factor by factor. The holomorphic hull of Remark 3.9.5 (i) exists as a minimal $\lambda \cdot \mathcal{O}$, and $-|\log(\underline{\Theta})| \in \mathbb{R}$ is obtained.

Lemma 7.4 (distortion of evaluation isomorphisms) [assembly]. The evaluation arrows of the diagram of Remark 3.10.2 (lower horizontal 1: [IUTchII] Corollary 4.10 (ii); upper horizontal 1: Corollary 4.6 (iv)(v)) do not shift degree functions, as prime-strip isomorphisms. Hence the total degree distortion of the IPL composite of Remark 3.11.1 (iii) is 1. Corollaries 4.5, 4.6, 4.10 are accepted as statements of the IUT main text. *Proof.* Corollary 4.10 (ii) states that the evaluation isomorphism of Corollary 4.6 (iv)(v) defines a prime-strip evaluation isomorphism $\mathfrak{F}^{\Vdash}_{\mathrm{env}} \xrightarrow{\sim} \mathfrak{F}^{\Vdash}_{\mathrm{gau}}$, and both sides have $\{\rho\}$ as constituents. By Lemma 3.2 the distortion is 1. Corollary 4.5 (v)

states that the global evaluation isomorphism is compatible with local evaluations under the Φ isomorphism, and the ρ^{Θ} formula of Example 3.5 (ii) fixes the local factors. The j^2 profile on the Gaussian side lies inside the definition of the objects. □

8 Same-run membership: chart calibration and the value-group switch

Lemma 8.1 (identity of chart calibration; not equality of profiles) [assembly]. The two routes (1-column log-Kummer after the link / a permutation isomorphism after 0-column log-Kummer) give the same calibration of $\Phi^{\mathrm{rlf}}_{\underline{v}} \cong \mathbb{R}_{\geq 0}$; that is, the positive constant multiple induced by the difference loop is 1. This lemma asserts neither that the Φ-profiles of the pilot elements of the two routes agree, nor same-run membership itself. *Proof.* Each arrow is a prime-strip isomorphism fixing the ρ-anchor $\log_{\Phi}(p_{\underline{v}})$, so the action of the difference loop of the two routes on $\Phi^{\mathrm{rlf}}_{\underline{v}}$ is an automorphism of the ordered topological monoid (a positive constant multiple) fixing the anchor, hence the identity. This says only that the two routes give the same ruler. In fact, at a bad prime the pilot profiles are, by route, $\{j^2 u_{\underline{v}}\}_j$ and $\{u_{\underline{v}}\}_j$, and the ratio of capsule averages is $\ell(\ell+1)/12 \neq 1$ ($\ell \geq 5$). Being measurable with the same ruler and the measured profiles being the same are different things; the latter does not follow from this lemma. □

Remaining hinge H*val (the value-group switch). The switch from Ψ_{LGP} to Ψ_q is compatible with the holomorphic hull, and from an actual output with a Gaussian/ Θ-side j^2 profile, an actual q-pilot representation belonging to the family of possible images is obtained. This is the core of strong (SHE)/(9-1) and does not follow from identity of chart calibration in Lemma 8.1. Theorem 9.1 of this paper includes H*val as an explicit hypothesis.

Lemma 8.2 (unit-side mutual compatibility from uniqueness characterizations) [assembly]. The three methods for cyclotomic rigidity isomorphisms are each characterized as the unique isomorphism preserving distinguished data, and agree on overlapping domains. Hence exhaustion on the unit side is treated. The value-group side H*val remains. Inputs: [EtTh] Theorem 1.6, Corollary 2.19; [IUTchI] Example 5.1 (v); [FrdII] Theorem 2.4; [AbsAnab] Proposition 1.2.1 (iv)(vi)(vii); [Mzk14] Theorem 6.12. *Proof.* [FrdII] Theorem 2.4 gives canonicity and functoriality of Kummer classes, but a ζ-scalar indeterminacy remains in principle in identifying fundamental classes (Remark 2.4.2). In the number-field method the isomorphism is unique by $\mathbb{Q}_{>0} \cap \widehat{\mathbb{Z}}^{\times} = \{1\}$ (Example 5.1 (v)); in the mono-theta method the valuation element $1 \in \widehat{\mathbb{Z}}$ and theta classes are preserved ([EtTh] Theorem 1.6 (ii)(iii)); in the MLF method it is fixed by Frobenius elements ([FrdII] Theorem 2.4 (ii)). The bridge is as follows. The distinguished data of the number-field method are the Kummer classes of actual functions obtained from divisors of κ-coric functions, and localization is restriction to decomposition groups. In the mono-theta method, the functorial algorithm of [EtTh] Corollary 2.19 (i) constructs the isomorphism, and Theorem 1.6 (iii) (together with the restriction algorithm of Corollary 2.19 (iii)) gives restriction compatibility of theta classes. Moreover, every isomorphism of local Galois groups preserves Frobenius elements, the cyclotomic character and residue maps ([AbsAnab] Proposition 1.2.1). Hence on overlapping domains the methods agree as the unique isomorphism preserving the Kummer classes of the same actual functions and the valuation element 1. □

9 Assembly of Corollary 3.12

Theorem 9.1 (conditional assembly) [assembly]. Under Semantic Principle S, Lemmas 3.2, 4.1, 4.2, 5.1, Proposition A, Lemmas 6.1, B, 7.1–7.4, 8.1, 8.2, the value-group switch **H*val** as an explicit hypothesis, and the known inputs of §12, $-|\log(\underline{\underline{\Theta}})| \geq -|\log(\underline{\underline{q}})|$ holds. *Proof.* If $-|\log(\underline{\underline{\Theta}})| \geq 0$ the claim is trivial, so assume it is negative. Take a concrete link image T_λ. By Lemma B, the explicit hypothesis H*val, and unit-side treatment in Lemma 8.2, the global representation of T_λ belongs to ${}^{1,\circ}\overline{\mathcal{U}}$ (corresponding to (IPL)(SHE) in the source). Compute the log-volume of T_λ in two ways. First, the value-group part of the link sends the Θ-pilot to the q-pilot (Step (i), Remark 3.8.1); by Proposition A the chart/transport distortion is $s_{\mathrm{chart}} = 1$, and by H*val the global representation reads as a q-pilot representation. Hence the aggregate defect is $\Delta = 0$ and the input-side reading equals $-|\log(\underline{\underline{q}})|$. Second, by the hull membership of Lemma B, the volume lies in $\mathbb{R}_{\leq -|\log(\underline{\underline{\Theta}})|}$. The two computations give $-|\log(\underline{\underline{q}})| \leq -|\log(\underline{\underline{\Theta}})|$. This is the conclusion of (xi-f) in the source, and $C_\Theta \geq -1$ follows formally. On the Lean side, sufficiency of this assembly is verified as `cor312_of_assembly`. □

10 Correspondence with LANA's (9-1)

The value-group switch H*val isolated in §8 is the central sentence corresponding to LANA's (9-1) ($\eta^q = \eta_S^{\mathrm{anab}}$). The point of the dictionary is as follows: the standard pilot of $\mathbb{R}^{\mathrm{val}}$ corresponds to the normalized generator of the value-group part, $\mathbb{R}^{\mathrm{ss}}$ to the set of log-volumes of possible images, the pilot of η_S^{anab} to the volume of the image T_λ of Lemma B, and the pilot of η^q to the volume of the input q-pilot. Proposition A gives that chart transport distortion is 1, but the equality does not follow from that alone. The equality follows, together with Lemma B, once H*val is accepted; this paper does not claim to have proved (9-1) itself. As a normalization caveat: LANA's q-pilot is in places built from a 2ℓ-th root of $q_{\underline{v}}$ and in places displayed as $\{q_{\underline{v}}\mathcal{O}_{\underline{v}}\}$; reading the statements as correspondences presupposes a fixed normalization ([W], §19). LANA's Lean asset `xi_iff` is a reciprocal parametrization of the defect equivalence here; symbols must not be identified letter by letter.

Relation to Mochizuki's "3.11.5". The "3.11.5" appearing in Figure 1 is not a remark number of IUT III but the name of an intermediate statement introduced in Mochizuki's formalization progress report [Form, §4], meaning Theorem 3.11 together with Remark 3.9.5; the report itself notes that this statement does not occur in [IUTchI–IV]. Read at the log-volume level, its (SHE) arrow is the reading in which the input data of the algorithm producing possible images is taken to be the q-pilot data constructed from étale Hodge theatres, i.e. the carrier reading of [U26], with locus $A \cap$ strong (SHE) $= A \cap K$. In relation to this paper: we do not posit "3.11.5" as a new hypothesis; §§7–8 expand the surrounding steps, including (Ob3)/(Ob4) of Remark 3.9.5 (vii). Since its core, the value-group switch, remains as H*val, however, this paper does not claim to have assembled "3.11.5" as a whole from the statement level, known literature and Principle S. The negative result of [U26] — inside the cited range A alone this part does not follow — remains.

Project LANA is formalizing from basic tools of anabelian geometry up to IUT, and there may be overlap with the author's work. The author regards this not as competition but as complementary: IUT verification becomes more robust when carried out by several people from different angles — audit of readings of the source, settling specifications by countermodels, formalization of the construction itself. This paper is independent work, not in a relation of coordination or endorsement with Project LANA.

The correspondence between (IPL)(SHE) in the source and this paper (the present reading) is as follows; where the correspondence lies, and the unresolved core, are made explicit in §§7–8.

Source notion	Correspondence here	Status
(IPL) input prime-strip link	Existence of the value-group-part isomorphism in Step (i) (Definitions 3.8, 2.4) and its degree distortion (Lemma 7.4)	Treated at statement level
Weak (SHE)	Output data are expressible in the target vocabulary (outer ellipse in Fig. 1)	Formalized in [U26]; alone it gives no conclusion
Strong (SHE) (= $A \cap K$)	The actual output family contains an actual q-pilot representation. Lemma 8.1 gives chart calibration, Lemma 8.2 the unit side; the value-group switch H*val itself is the central remainder; in Lean it is the pair of hypotheses `hmem` (membership of a single global object) and `hlocal` (corrected place-wise equality, which at a bad place is the label-wise profile equality)	Surroundings treated; H*val (R2V) unresolved, in-situ type transfer is R2
(Ob4)	Lemma 7.2 (well-definedness of assembly)	Treated at statement level

11 The height inequality and the state of constants

The output of the assembly has been connected to a height inequality of the shape of IUT IV Theorem 1.10 ([W], §§8, 14). The state of constants is fixed up to $\eta_{\mathrm{prm}} = 17$, $\xi_{\mathrm{prm}} = e^{46}$, the enumeration of four exceptional values of j, and the decomposition $T_{\mathcal{K}} = B^*_{\mathcal{K}} + 2.482$. As an example of individual numerical computation, exact integer computation for four one-parameter families over $\mathbb{Q}$ ($t = 2, \ldots, 200$) and a finite family (399 curves) extracted as semistable and minimal from 11a1 and 37a1 gives, as a maximum inside this finite family, $\max(h(j_E) - h)/12 \approx 0.852886$ (attained at $t = 7$ in family F4). This finite-family maximum is not a uniform bound over $\mathcal{K}_V$ as a whole; for the whole family we refer to the loose bound about 332.1 given by the formula of [W], §20.2. Reference values obtained by substituting the finite-family maximum are $T_{\mathcal{K}} \approx 3.334886$ and $H_{\mathcal{K}} \geq 67.856 + 8B^*_{\mathcal{K}} \approx 74.679$. The coefficient condition of [W] Theorem 8.1 (in particular prime $\ell \geq 7$) is also made explicit. The explicit-constant inputs are Broadbent–Kadiri–Lumley–Ng–Wilk, Löbrich (via Gaudron–Rémond), and Sijsling.

12 The KNOWN ledger

This is the list of results quoted here as black boxes. Status: [V] verified in the source; [E] quoted as established literature (the author's policy for Mochizuki's papers other than IUT I–IV); [C] citation level; [Cl] classical standard results. Numbered results of IUT I–IV themselves are not listed here and are used as black boxes throughout. Places of use are given by sections and lemmas of this paper; where a number of the detailed draft [W] is used it is marked [W]. Details of the identification of exceptional j in §11 are delegated to [W], §20, where [CanLift] Proposition 2.7 and [Take2] Theorem 4.1 (i) are used as pointers ([Take2] itself has not been verbatim-checked).

Input	Content and use	Status	Source
KNOWN-1	Frobenius normalization (numerical fixing preserving generators). §§4–5 here	[V]	[AbsAnab] Prop. 1.2.1 (iv) (same normalization in [AbsTopIII] Cor. 1.10 (b))
KNOWN-2	Single-scalarity (via Dirichlet's unit theorem). Globalization of Proposition A in §5 here; [W] Lemma 13.2	[E]	[FrdI] Thm. 6.4 (i)(ii) (statement verbatim-checked)
—	The non-tensor-power version determines the tensor-power version via the naive Frobenius functor, degrees multiplied uniformly by M. Lemma 7.2 here	[E]	[FrdI] Prop. 2.1 (i)(ii)
—	Places relevant local Frobenioids in the model class, making Thm. 6.4 applicable. Lemma 7.2 here	[E]	[FrdI] Thm. 5.2 (ii)
KNOWN-3	Cyclotomic, discrete and constant-multiple rigidity in mono-theta environments. Lemmas B, 8.2 here	[E]	[EtTh] Thms. 1.6, 1.10, Cors. 2.18, 2.19 (proofs depend on [Mzk14] §6)
—	Canonicity/functoriality of Kummer classes and locus of the scalar indeterminacy in fundamental-class identification. Lemma 8.2 here	[E]	[FrdII] Thm. 2.4, Rem. 2.4.2
—	Temp-absolute preservation of decomposition groups, standard integral structures, torsion points	[E]	[Mzk14] Thms. 6.5 (iii), 6.8, 6.12, Cor. 6.11
—	Definition of log-volume and log-volume formula for mono-analytic integral structures. §5 here; [W] Lemma 3.4	[E]	[AbsTopIII] Props. 5.7, 5.8 (iii), Cor. 5.10 (ii)
—	Complete determination of Kummer isomorphisms (γ' anchor 4). Lemma 6.1 here	[V]	[FrdII] Def. 2.1 (ii) (verbatim-checked)
KNOWN-4	Effectivity of η_{prm}, ξ_{prm} by explicit prime estimates. [W] Thm. 8.1	[Cl]	Rosser–Schoenfeld type estimates (outside the text)
—	SL(2) structure and isogeny estimates for elliptic curves in general position. (P4)(P6) of [W] Cor. 2.2	[V]	[GenEll] Thm. 2.1, Lems. 3.1, 3.5
—	Faltings height comparison. Final step via [W] Cor. 2.2	[C]	Silverman [12][13] (Cornell–Silverman 1986); not in hand, not verbatim-checked
—	Classical theta-function formulae	[Cl]	[Mumf] pp. 306–307
—	Explicit prime estimates (effectivity of ξ_{prm}). [W] Prop. 14.2	[Cl]	Broadbent–Kadiri–Lumley–Ng–Wilk (arXiv:2002.11068) Thm. 1
—	Explicit Faltings-height constants (alternative route to Silverman [12][13]). [W] §18	[Cl]	Löbrich (arXiv:1505.00602) Props. 3.1, 3.2 (citing Gaudron–Rémond)

Input	Content and use	Status	Source
—	Enumeration of four exceptional values of j. [W] §20	[Cl]	Sijsling (arXiv:1707.01158), Table 4

13 What remains: list of inputs and non-claims

Input	Content	Status
IUT black box	Numbered statements of IUT I–IV (including Theorem 3.11) quoted as correct	Unverified (outside scope by design)
Value-group switch H*val (R2V)	Hull compatibility of the switch $\Psi_{\mathrm{LGP}} \to \Psi_q$; does not follow from chart calibration (Lemma 8.1) or unit-side treatment (Lemma 8.2)	Unresolved; core of strong (SHE)/(9-1)
Semantic Principle S	Prime-strip isomorphisms intertwine all constituents; compatibility as commutative squares (Lemmas 3.2, 6.1 here; [W] Lemma 11.1, γ')	Expanded to definition level; justification itself not discussed
Known literature [E]	[FrdI], [FrdII], [EtTh], [AbsTopIII], [AbsAnab], [Mzk14], etc. (Mochizuki's non-IUT papers)	Quoted as established literature (author's policy)
Classical inputs	Dirichlet's unit theorem, local class field theory / Brauer groups, uniqueness of Haar measure, external explicit-constant results	Quoted as standard results
Frame fidelity (R2)	The squares, non-degeneracy and local cancellation checked in Lean model pairs and the type skeleton hold, for the whole composites, on the same generators and same pilot objects of the actual construction	Unfinished; precise transfer problem of §3.1
Independent re-verification	Third-party line-by-line re-reading of the working draft [W] itself	Not yet done

The history of external cross-checking is also recorded. Early versions of this work ([W] v0.6–v0.8) described hinges H1, H2 as "dissolved", but independent external checking pointed out that the opening sentence of [W] Lemma 13.2 asserted more than its proof, that identity of the value-group part formed a deferral loop, and that the object-level sentence of [W] Lemma 3.4 (a) was wrong; all were confirmed correct as criticisms by citation checking. Moreover, by an external adjudication of 12 November 2026, the assertion "the Φ-components agree" in Lemma 8.1 here (the passage corresponding to [W] Lemma 13.5) was withdrawn and downgraded to identity of chart calibration. Lemma 8.2 remains as the unit side, but the value-group switch H*val is unresolved. The ratings in this paper are after these corrections. The centre of the remaining work is the value-group switch H*val itself and inhabitation of Principle S in the actual construction; together with verification of Theorem 3.11 itself, this note can also be read as a specification of statements to be supplied by continuing projects, such as Project LANA, that formalize the IUT construction as a whole. Remaining work completable on this side includes individual numerical computation of

$B^*_{\mathcal{K}}$, revision of the corresponding passage of the detailed draft [W], and independent re-verification of the drafts.

Disclosure

This paper is the author's personal work as an independent researcher, unrelated to any institutional affiliation. Generative AI was used, under the author's direction, in the proofs, assembly, audit, formalization, and preparation of the Japanese and English drafts of this paper. The author bears responsibility for the contents. The author's own complete reading of the sources is in progress, and this paper is a working report not yet independently re-verified.

Artifact availability

The Lean capstone `cor312_of_assembly` (§2, §9), the arithmetic inhabitation and the prime-strip skeleton / Tate model-pair files referred to in §3.1 are reproduced for reference in the supplementary archive. They are excerpts of the author's larger Lean development and are not standalone: the selected build closing over them (353 modules) was checked in that development. The Lean check of Figure 1, `VennRegionB.lean`, is included as well; it imports two modules of the same development (the hull operator and the profile countermodel of [U26c]), restates the capstone in the same shape, and was compiled against them with Lean 4.30.0 and the same Mathlib pin. What is machine-checked in this paper is sufficiency of the assembly and the model-level statements named in the text — not the paper proof as a whole, and not frame fidelity (R2) or the value-group switch (R2V). The TeX source of this paper, the supplementary files and checksums are at the GitHub tag `cor312-assembly-v1.0.3`.

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